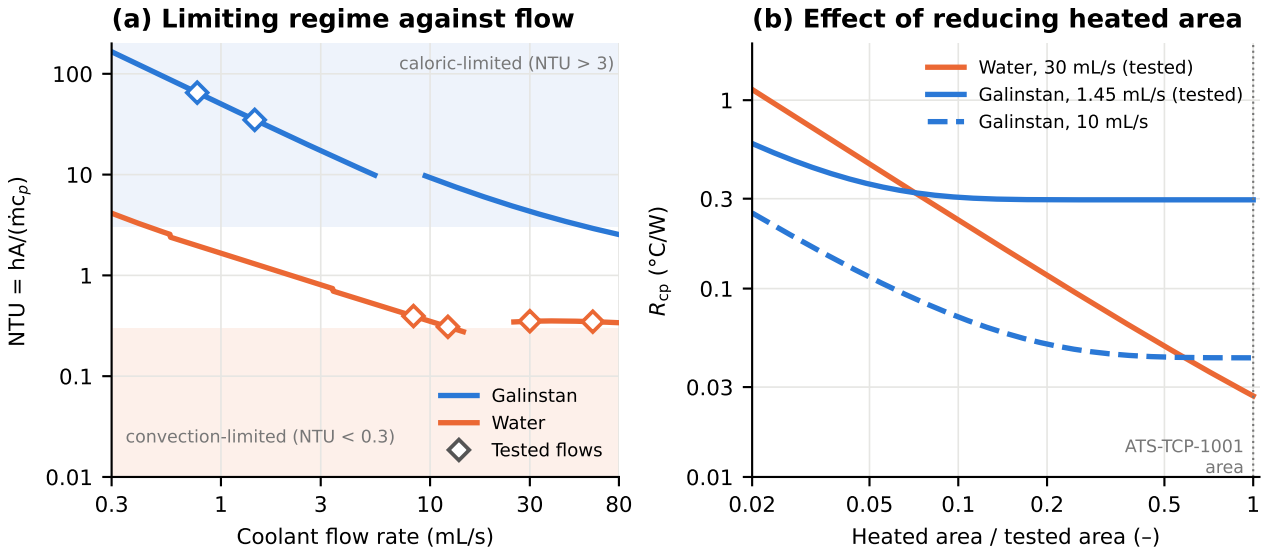
(a) Limiting regime against flow
caloric-limited (NTU > 3)
convection-limited (NTU < 0.3)
Galinstan
Water
Tested flows
NTU = hA/(ṁc$_p$)
100
10
1
0.1
0.01
0.3
1
3
10
30
80
Coolant flow rate (mL/s)
(b) Effect of reducing heated area
Water, 30 mL/s (tested)
Galinstan, 1.45 mL/s (tested)
Galinstan, 10 mL/s
ATS-TCP-1001
area
R_{cp} (°C/W)
1
0.3
0.1
0.03
0.01
0.02
0.05
0.1
0.2
0.5
1
Heated area / tested area (–)